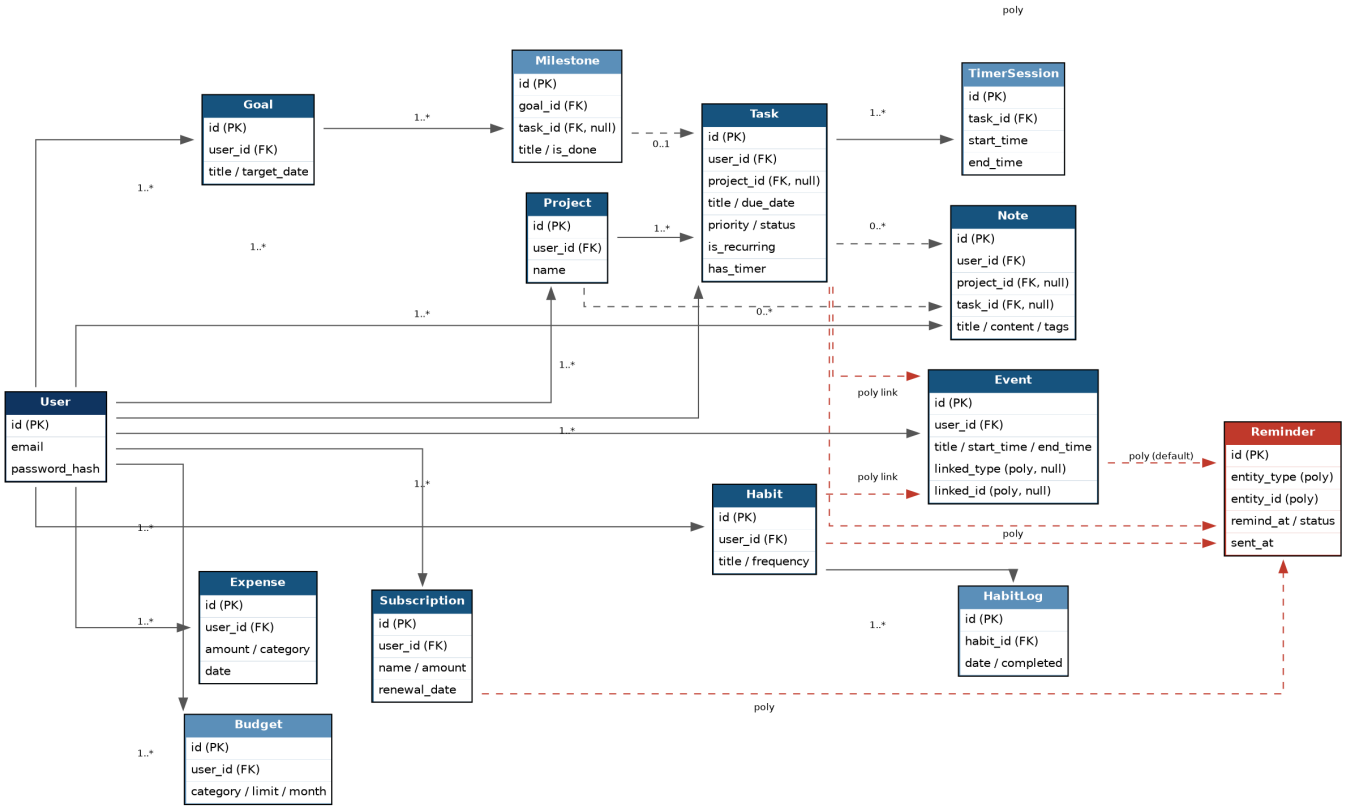
Life Style App

Entity-Relationship Diagram (ERD) - full data model

This covers every table needed for the MVP, based on all decisions made (module scope + architecture). Dashed red lines represent polymorphic relationships (Reminders, and Event/Task/Habit links) that are resolved in application code rather than as real database foreign keys.



Full Table Definitions

User

Column	Type	Notes
id	PK	
email	string, unique	
password_hash	string	Django's built-in hashing
created_at	datetime	

Project

Column	Type	Notes
id	PK	
user_id	FK -> User	
name	string	
description	text (nullable)	
created_at	datetime	

Task

Column	Type	Notes
id	PK	
user_id	FK -> User	
project_id	FK -> Project (nullable)	
title / description	string / text	
due_date	datetime (nullable)	
priority	enum	low / medium / high
status	enum	todo / in_progress / done
is_recurring	boolean	+ recurrence_rule if true
has_timer	boolean	enables TimerSession tracking
created_at	datetime	

TimerSession

Column	Type	Notes
id	PK	
task_id	FK -> Task	
start_time	datetime	

end_time	datetime (nullable)	duration computed at query time

Habit

Column	Type	Notes
id	PK	
user_id	FK -> User	
title	string	
frequency	enum	daily / weekly / custom
created_at	datetime	

HabitLog

Column	Type	Notes
id	PK	
habit_id	FK -> Habit	
date	date	
completed	boolean	one row per day, used for streaks

Goal

Column	Type	Notes
id	PK	
user_id	FK -> User	
title / description	string / text	
target_date	date (nullable)	

Milestone

Column	Type	Notes
id	PK	
goal_id	FK -> Goal	
task_id	FK -> Task (nullable)	optional link to an actual task
title	string	
is_done	boolean	

Note

Column	Type	Notes
id	PK	

user_id	FK -> User	
project_id	FK -> Project (nullable)	
task_id	FK -> Task (nullable)	
title / content	string / text	
tags	array/string	
created_at	datetime	

Expense

Column	Type	Notes
id	PK	
user_id	FK -> User	
amount	decimal	
category	string	
date	date	
note	string (nullable)	

Subscription

Column	Type	Notes
id	PK	
user_id	FK -> User	
name	string	
amount	decimal	
renewal_date	date	
billing_cycle	enum	monthly / yearly

Budget

Column	Type	Notes
id	PK	
user_id	FK -> User	
category	string	
monthly_limit	decimal	
month	date	first day of the month, for scoping

Event

Column	Type	Notes
id	PK	
user_id	FK -> User	
title	string	
start_time / end_time	datetime	
linked_type	string (nullable, poly)	"task", "habit", or null
linked_id	poly ref (nullable)	resolved in app code
all_day	boolean	

Reminder

Column	Type	Notes
id	PK	
entity_type	string (poly)	"task" / "habit" / "subscription" / "event"
entity_id	poly ref	resolved in app code
remind_at	datetime	
offset_description	string (nullable)	e.g. "15 minutes before"
status	enum	pending / sent / dismissed
sent_at	datetime (nullable)	
created_at	datetime	

Key Relationship Notes

- Every top-level entity (Project, Task, Habit, Goal, Note, Expense, Subscription, Budget, Event) belongs to exactly one User.
- Milestone -> Task is optional (0..1): a milestone can be tracked purely as a checklist item, or backed by a real Task.
- Note can optionally link to a Project or a Task (or neither) - it is never required to have a parent.
- Event.linked_type/linked_id is polymorphic and optional: an event can be fully standalone (e.g. a lecture) or tied to a Task/Habit.
- Reminder.entity_type/entity_id is polymorphic and can point to a Task, Habit, Subscription, or Event - this is the single shared table for all reminder delivery, run server-side via Celery/Celery Beat.
- Creating an Event triggers an automatic Reminder (start_time - 15 min) in the same atomic transaction.
- Files/attachments are intentionally left out of this ERD version - Note and Project already have the shape needed to add an Attachment table later without breaking existing relationships.

Polymorphic fields (entity_id, linked_id) are not enforced as real foreign keys by the database - validity must be checked in Django application/service logic before writing to these tables.